

```

def find(parent, i):
    if parent[i] == i:
        return i
    return find(parent, parent[i])

def union(parent, a, b):
    root_a = find(parent, a)
    root_b = find(parent, b)
    parent[root_a] = root_b

def kruskal(vertices, edges):

    edges.sort(key=lambda x: x[2])

    parent = []

    for i in range(vertices):
        parent.append(i)

    mst = []
    total = 0

    for u, v, weight in edges:

        root_u = find(parent, u)
        root_v = find(parent, v)

        if root_u != root_v:

            mst.append((u, v, weight))
            total += weight
            union(parent, root_u, root_v)

    return mst, total

vertices = 4

edges = [
    (0, 1, 10),
    (0, 2, 6),
    (0, 3, 5),
    (1, 3, 15),
    (2, 3, 4)
]

mst, total = kruskal(vertices, edges)

print("Edges in Minimum Spanning Tree:")

```

```

Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
==== RESTART: C:/Users/ROOPA SAI DIVYA/OneDrive/Documents/Big-O, Ω and Θ.py ====
Edges in Minimum Spanning Tree:
2 - 3 = 4
0 - 3 = 5
0 - 1 = 10
Minimum Cost = 19

>>>

```